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RESEARCH

Geospatial analysis of drought susceptibility in Marathwada region, Maharashtra, India

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Abstract

Marathwada, a semi-arid region in Maharashtra, India, frequently experiences droughts that severely impact agriculture and water resources. This study developed a drought susceptibility map using a spatial multicriteria decision-making framework based on the Analytic Hierarchy Process (AHP). Six key parameters: rainfall anomaly, Soil Moisture Anomaly Percentage Index (SMAPI), Normalized Difference Vegetation Index (NDVI), Vegetation Condition Index (VCI), Vegetation Health Index (VHI), and Temperature Condition Index (TCI), were integrated due to their proven relevance in assessing meteorological and agricultural droughts in semi-arid climates. Assigned weights were: VHI (28.42%), rainfall anomaly (27.44%), SMAPI (25.59%), VCI (8.97%), NDVI (6.47%), and TCI (3.11%). The resulting drought susceptibility map categorized the region into five classes: very low, low, moderate, severe, and extreme. Districts such as Beed, Osmanabad, and Latur were identified as high-risk zones, marked by rainfall anomalies below −230 mm, soil moisture below 36%, and vegetation stress. In contrast, districts like Hingoli and eastern Nanded exhibited high NDVI (>0.35), VHI (>63%), and SMAPI (>68%), placing them in low-susceptibility zones. This spatial assessment highlights the utility of combining climatic, thermal, and vegetation indices for drought risk evaluation and provides a valuable tool for targeted mitigation, agricultural planning, and adaptive water resource management by local and regional authorities.

1 Introduction

Agricultural drought is one of the most critical natural hazards affecting global food security, ecosystem balance, and rural livelihoods. It occurs when soil moisture becomes inadequate to sustain crop growth during the critical stages of development, leading to yield reduction and economic losses (Rahman et al. 2025; Singh et al. 2025). Globally, drought impacts nearly half of the earth's surface annually and causes billions of dollars in agricultural losses (Kogan et al. 2013; Orimoloye et al. 2022). The severity of drought has intensified due to climate change, increased frequency of El Niño events, deforestation, and over-exploitation of groundwater (Pandit 2024; Jiang et al. 2025). Countries such as Ethiopia, Australia, United States, and China have experienced significant agricultural setbacks due to recurring droughts(Miyan 2015; Rahman et al. 2025). The socio-economic consequences include food insecurity, forced migration, and rural poverty (Zaini et al. 2019). Advanced research has highlighted the need for real-time drought monitoring systems to help mitigate these effects (Ahmed Osmanr et al. 2025). Integrated approaches using meteorological, hydrological, and socio-economic parameters are being promoted for robust drought assessment (Rahman et al. 2025). New methodologies now utilize indices like the Standardized Precipitation Index (SPI), Vegetation Health Index (VHI), and Joint Drought Index (JDI) to measure and classify drought severity across landscapes (Bageshree et al. 2022). Technological interventions like remote sensing, Geographic Information Systems (GIS), and machine learning have revolutionized drought risk analysis on global, regional, and local scales (Maity et al. 2024). However, despite significant progress in global drought science, localized studies remain crucial for designing region-specific adaptation and mitigation strategies (Wang and Ren 2025).

India, with its vast agro-climatic diversity and dependence on monsoon rains, is highly vulnerable to droughts. Agricultural drought in India affects more than 70% of its rural population, particularly in semi-arid and dry

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sub-humid zones (Niranjan Kumar et al. 2013; Todmal 2022). States like Maharashtra, Karnataka, Rajasthan, Tamil Nadu, and Andhra Pradesh have recurrently faced drought crises that have jeopardized agricultural productivity and rural stability (Amrit et al. 2020; Kannan and Govindasamy 2022; Kannan et al. 2025). According to the Economic Survey of Maharashtra, erratic monsoons have intensified the variability in crop production and groundwater recharge(GoM 2022). Marathwada and Vidarbha regions, forming the heart of Maharashtra's agricultural belt, are frequently impacted by both meteorological and agricultural droughts (Kelkar et al. 2020). Despite normal rainfall, these areas suffer from localized dry spells, poor water management, and heavy dependence on rain-fed farming, making drought assessment more complex (Ganjir et al. 2025). Efforts by the Indian government and scientific community have led to the development of systems like MAHA-MADAT and crop insurance schemes, but real-time drought tracking remains a challenge (Aswathi et al. 2018). Remote sensing has become instrumental in assessing drought over large regions with limited ground data. Studies in India have successfully combined satellite-based indicators with rainfall and evapotranspiration models to assess drought vulnerability (Pandey et al. 2012). Socio-economic indices like the Composite Agricultural Vulnerability Index (CAVI) have also been integrated with biophysical data to identify highly exposed zones (Palchaudhuri and Biswas 2016). However, to address the spatial heterogeneity of drought, region-specific assessments based on localized parameters remain indispensable for sustainable agricultural planning.

The use of remote sensing and satellite data has transformed agricultural drought monitoring by providing timely, consistent, and spatially detailed insights. Satellite sensors such as MODIS, Landsat, and Sentinel enable multi-spectral analysis to track vegetation health, surface temperature, and moisture anomalies (Kogan et al. 2013). Drought indices derived from satellite imagery such as NDVI, VCI, TCI, and particularly the Vegetation Health Index (VHI), have proven effective in identifying drought onset, severity, and recovery patterns (Susila et al. 2024). These indices allow real-time evaluation of vegetation stress during critical crop stages. NDVI responds to chlorophyll content, VCI normalizes vegetation condition across years, and TCI evaluates temperature-related vegetation stress together forming a comprehensive VHI (Mullapudi et al. 2023). Unlike rainfall-only assessments, satellite-based monitoring incorporates spatial patterns of crop health and land surface changes, which is critical in fragmented farmlands and dryland regions (Gaikwad et al. 2015). Advanced geoinformatics techniques using AHP-GIS and machine learning models (e.g., Random Forest and XGBoost) have been applied to integrate multiple datasets such as rainfall, LST, NDWI, and SMI into composite drought risk maps (Alharbi et al. 2022; Ahmed Osmanr et al. 2025). However, recent studies emphasize the importance of clearly distinguishing between drought susceptibility and drought severity, particularly when using multi-year indicators to assess spatial vulnerability rather than event-specific intensity (Guria et al. 2025). While the AHP-GIS framework offers a structured and transparent decision-making tool, recent research suggests that machine learning approaches often yield improved predictive accuracy and robustness in drought susceptibility modelling, especially under data heterogeneity and non-linear interactions (Senapati & Das, 2025c). Moreover, integrating early warning systems and forecasting models can enhance operational relevance and drought preparedness (Senapati et al. 2021, 2025a). Remote sensing is particularly effective in data-scarce and cloud-prone regions, improving early warning systems and mitigation planning (Hamidi et al. 2025). Its compatibility with decision-support frameworks ensures rapid policy formulation and disaster response. The synergy between remote sensing, field data, and participatory tools promises a transformative leap in drought risk management and agricultural sustainability (Azmir 2025).

The Marathwada region of Maharashtra, comprising eight districts, lies in the rain-shadow zone of the Western Ghats and is one of India's most drought-prone regions. Its agriculture is heavily dependent on the southwest monsoon, with minimal irrigation coverage and high climatic variability. The frequent occurrence of droughts has led to food insecurity, farmer indebtedness, and agrarian distress (Guhathakurta and Saji 2013; Todmal 2019; Amrit et al. 2020). The primary objective of this study is to assess and map the spatial extent and severity of agricultural droughts in Marathwada using rainfall records (1993–2023) and remote sensing indices. The study aims to identify drought-vulnerable zones using the Analytic Hierarchy Process (AHP), incorporating multiple physical and socio-economic indicators. This region-specific mapping will enable better targeting of government interventions, climate-resilient farming practices, and drought preparedness strategies. Given the pressing water scarcity and agrarian challenges, such localized drought assessments are critical for promoting adaptive agricultural management and improving rural livelihoods. The novelty of this study lies in its integration of a 31-year temporal dataset (1993–2023) with six indices, such as rainfall anomaly, SMAPI, NDVI, VCI, VHI, and TCI, using an AHP-based spatial modelling framework specifically designed for the Marathwada region. This combination of meteorological, thermal, vegetation, and soil moisture

parameters offers a robust multi-dimensional perspective of drought vulnerability. Unlike many previous studies focusing on short-term or single-index approaches, the current model emphasizes regional-scale operational applicability, enabling targeted decision-making for drought-prone districts in India.

2 Study area

Marathwada lies in the central-eastern part of Maharashtra, India, between 17°37′N to 20°39′N latitude and 74°33′E to 78°22′E longitude, covering an approximate area of 64,813 sq. km (Fig. 1). Geographically, it is part of the Deccan Plateau and falls in the rain-shadow region of the Western Ghats, resulting in a semi-arid climate with high inter-annual rainfall variability (Amrit et al. 2020). The region receives 500–750 mm of rainfall annually, mostly during the southwest monsoon (June to September), but with erratic distribution and frequent dry spells, which contribute to recurring droughts (Todmal 2019). Vegetation in the region consists of degraded dry deciduous forests, with common species like Babul, Acacia, and Neem, although much of the natural cover has been cleared for agriculture (Gaikwad et al. 2015). Marathwada's topography is gently undulating with some basaltic hill ranges, and elevation varies between 300 and 600 m above sea level, impacting local microclimates and water retention (Kelkar et al. 2020). Soils are predominantly black cotton (Regur) soils derived from Deccan Trap basalt, known for their high moisture retention but poor drainage, making them highly sensitive to rainfall variability (Ganjir et al. 2025). Agriculture is the mainstay, with over 70% of the population engaged in farming, primarily reliant on rain-fed conditions; major crops include jowar, bajra, soybean, cotton, tur, and sugarcane (Guhathakurta and Saji 2013; Mullapudi et al. 2023). The region is drained by several tributaries of the Godavari River, such as the Manjira, Purna, and Penganga. Although overextraction and lack of conservation infrastructure have led to widespread groundwater depletion (Ganjir et al. 2025). The districts are well-connected through national highways like NH-52, NH-61, and NH-63, facilitating regional trade and access to input and output markets. Each of Marathwada's eight districts—Aurangabad, Beed, Latur, Nanded, Osmanabad, Parbhani, Jalna, and Hingoli has its own unique agro-climatic profile and drought exposure levels. Climatic conditions are marked by high summer temperatures (often exceeding 42 °C), low humidity, and high potential evapotranspiration, which together accelerate soil moisture loss and intensify drought risk (Kelkar et al. 2020).Socio-economically, the region faces high rural poverty, small and marginal landholdings, poor irrigation infrastructure, and significant agrarian distress due to frequent crop failures (GoM 2022; Bageshree et al. 2022). Aurangabad serves as the industrial and administrative hub; Beed and Osmanabad face frequent droughts and high groundwater exploitation and Latur has faced severe drinking water crises in the past (Mullapudi et al. 2023). Districts like Parbhani and Jalna contribute significantly to research and seed production, while Nanded benefits from the Godavari River but still suffers from spatial rainfall inequality (Bageshree et al. 2022; Susila et al. 2024).

3 Materials and methods

3.1 Data used

A wide range of spatial and temporal datasets were used for the assessment and mapping of agricultural drought in the Marathwada region of Maharashtra over a 31-year period (1993–2023) (Table 1). The datasets were chosen for their resolution, availability, and suitability for monitoring key drought indicators such as rainfall variability, vegetation health, temperature stress, and soil moisture. As the input datasets varied widely in spatial resolution from high-resolution Sentinel-2 imagery (10 m) to coarse-resolution GLDAS soil moisture data (25 km), a spatial harmonization process was applied. All layers were resampled to a common resolution (1 km) using bilinear interpolation for continuous data (rainfall anomaly, SMAPI, LST) and nearest-neighbour for categorical datasets (LULC). This harmonization minimized scale mismatches and allowed consistent integration of datasets in the AHP-GIS model.

The methodological framework is designed to integrate remote sensing data, climatic datasets, and spatial decision modelling through AHP to generate a Drought Susceptibility Map (DSM) (Fig. 2).

The process was carried out in sequential phases: (i) drought year identification through precipitation anomaly analysis, (ii) computation of remote sensing indices, (iii) spatial normalization of drought parameters, and (iv) modelling and generation of the susceptibility map.

3.2 Rainfall-based drought analysis

3.2.1 Precipitation raster generation

Rainfall data from 1993 to 2023 were used to generate monthly rainfall raster, seasonal rainfall raster and annual rainfall raster.

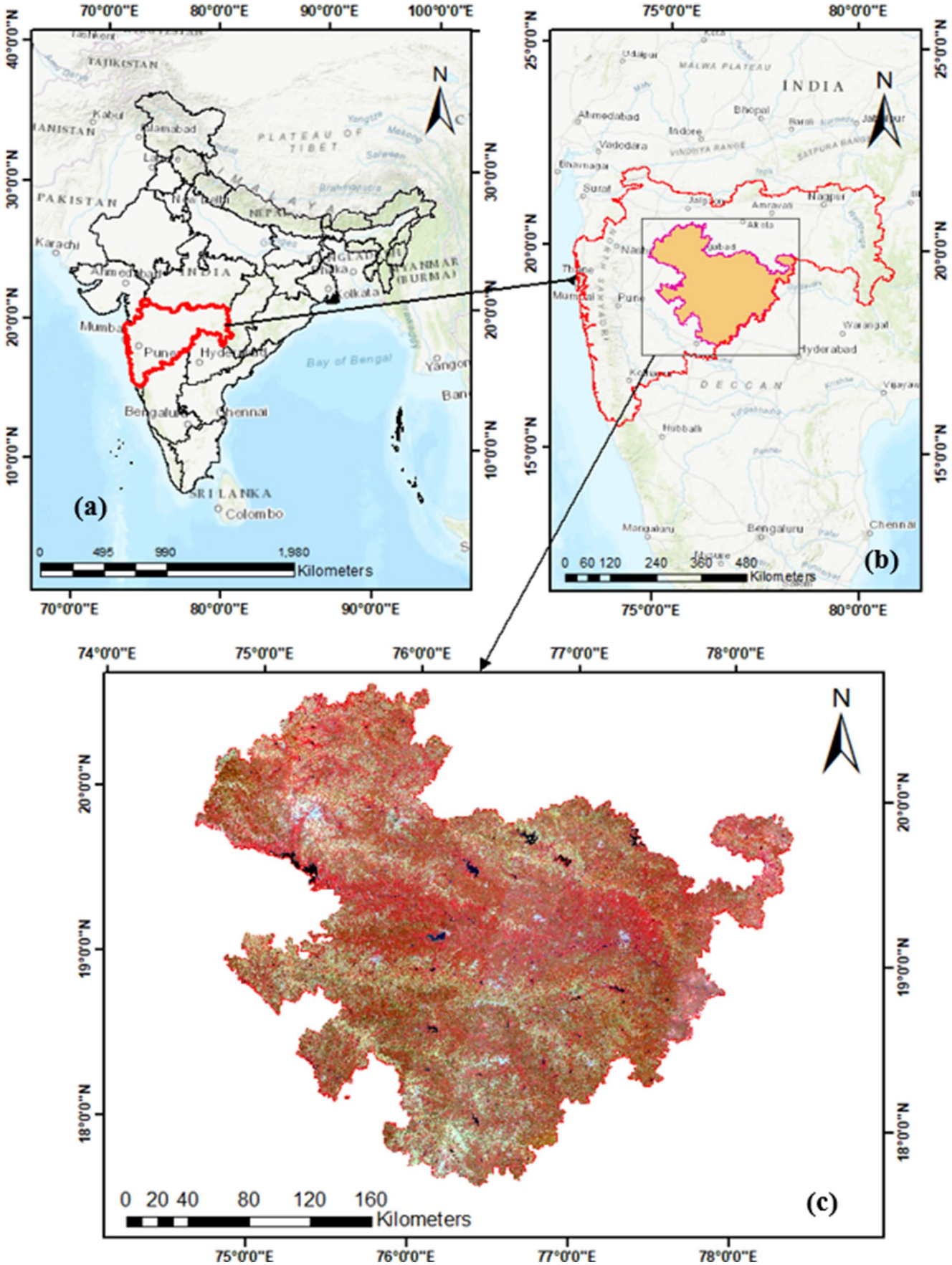

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Fig. 1 The figure illustrates (**a**) India with the state of Maharashtra, **b** Maharashtra state with the Marathwada region, and (**c**) a high-resolution FCC satellite image of Marathwada

3.2.2 Long-term mean computation

The 31-year climatological mean was computed to establish the baseline rainfall conditions using the following formula:

$$\bar{R}_{mean} = \frac{1}{N}\sum_{i=1}^{n} R_i \quad (1)$$

where R_i is rainfall for year i and N is 31-years.

3.2.3 Rainfall anomaly and deviation

To quantify interannual rainfall variability and detect anomalous drought patterns, two key metrics were calculated based on deviations from the long-term climatological mean:

Rainfall anomaly:

$$A = R_{year} - \bar{R}_{mean} \quad \ldots\ldots\ldots\ldots (2)$$

Rainfall deviation (%):

$$D = \frac{R_{year} - \bar{R}_{mean}}{\bar{R}_{mean}} \times 100 \quad (3)$$

Table 1 Summary of spatial and temporal datasets used for drought assessment in the Marathwada region

Dataset	Spatial resolution	Purpose	Source
Sentinel-2	10 m	FCC & Cropland identification	Copernicus Data Space
Landsat 8 (C2 Level 2 SR)	30 m	NDVI (2014–2023)	USGS Earth-Explorer
MODIS MOD13Q1	250 m	Median NDVI (2000–2013)	NASA LP DAAC
MODIS LST	1 km	Temperature Condition Index (TCI)	NASA LP DAAC
LULC (Sentinel-2)	10 m	Cropland mask	Copernicus Data Space
LULC (MODIS MCD12Q1 V6)	500 m	Cropland identification	NASA LP DAAC
TerraClimate	4 km	Rainfall anomaly, seasonal means	Climate Engine
GLDAS NOAH v2.1	25 km	Soil moisture anomaly	NASA Earthdata
SRTM DEM	30 m	Terrain slope	USGS Earth-Explorer

These metrics enabled identification of significant drought years and supported further spatial-temporal analysis (Mishra and Singh 2010).

3.3 Remote sensing-based drought indices

Vegetative and thermal drought conditions were assessed using the following satellite-derived indices:

3.3.1 Normalized difference vegetation index (NDVI)

To evaluate vegetation health across the study area, the NDVI was computed using red and near-infrared spectral reflectance values:

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (4)$$

This index reflects vegetation health and chlorophyll activity. Higher values indicate dense, healthy vegetation (Essaadia et al. 2022).

3.3.2 Vegetation condition index (VCI)

To assess vegetation stress relative to historical conditions, VCI was derived by normalizing current NDVI values between their historical maximum and minimum:

$$VCI = \frac{NDVI_{current} - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \times 100 \quad (5)$$

VCI normalizes NDVI between its historical minimum and maximum to represent vegetation stress.

3.3.3 Temperature condition index (TCI)

To evaluate thermal stress on vegetation, TCI was calculated by comparing current land surface temperature to its historical extreme values:

$$TCI = \frac{LST_{max} - LST_{current}}{LST_{max} - LST_{min}} \times 100 \quad (6)$$

TCI measures temperature-induced vegetation stress using MODIS LST data.

3.3.4 Vegetation Health Index (VHI)

To integrate both vegetative and thermal stress into a single index, VHI was calculated as a weighted combination of VCI and TCI:

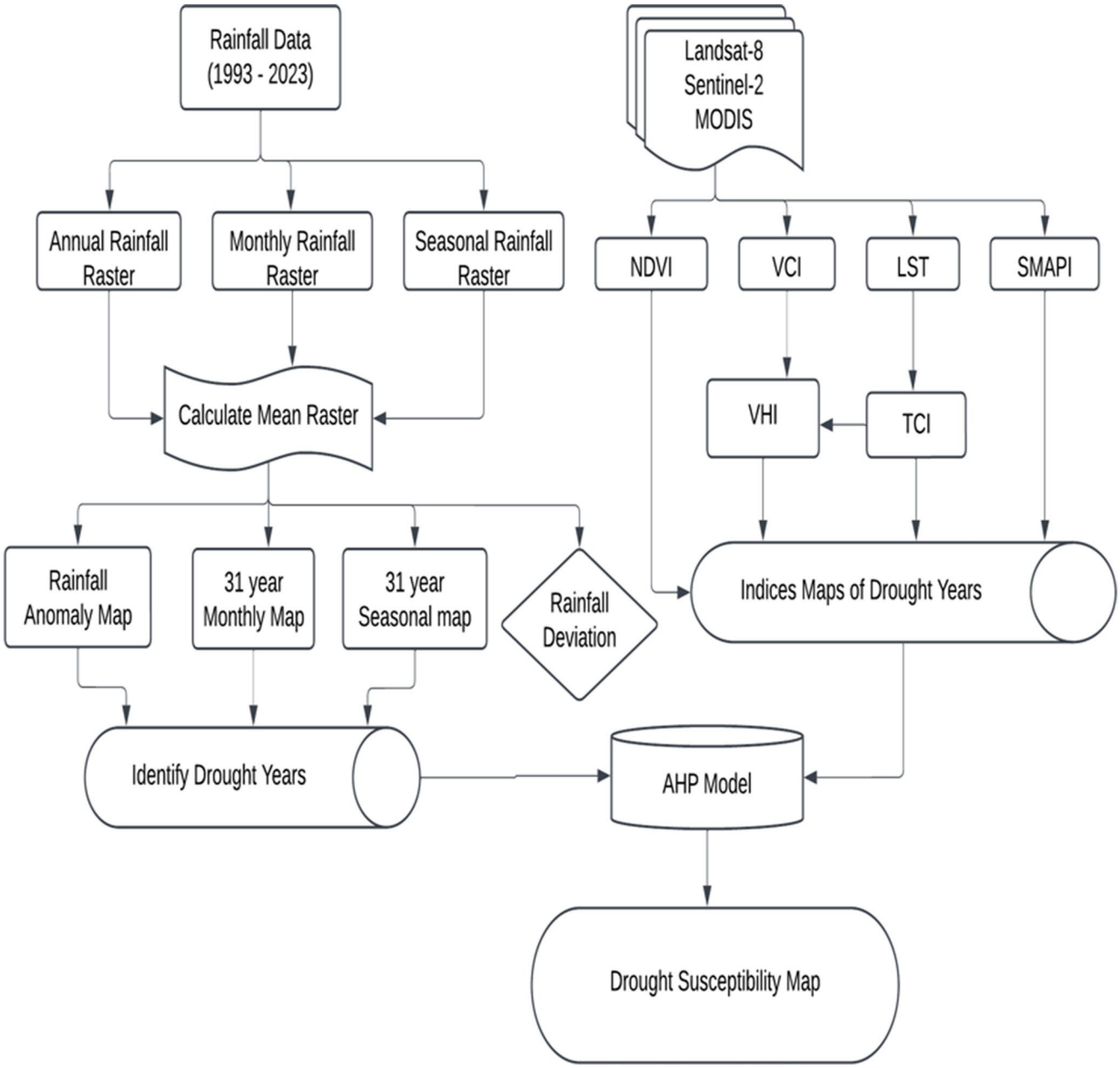


Fig. 2 Flowchart illustrating the methodology adopted for drought susceptibility mapping in Marathwada (1993–2023)

$$VHI = \alpha \cdot VCI + (1 - \alpha) \cdot TCI \tag{7}$$

where $\alpha = 0.5$. VHI combines VCI and TCI to provide a comprehensive index of vegetative and thermal drought (Kogan et al. 2013).

3.3.5 Soil moisture anomaly percentage index (SMAPI)

To estimate root-zone water stress, SMAPI was computed by assessing deviations in current soil moisture from historical averages:

$$SMAPI = \frac{SM_{current} - SM_{mean}}{SM_{min}} \times 100 \tag{8}$$

SMAPI helps to evaluate root-zone water stress, crucial for rainfed agriculture (Ding et al. 2023). These indices were spatially processed for each identified drought year.

3.4 Analytical Hierarchy Process (AHP)

The Analytic Hierarchy Process (AHP) was chosen due to its ability to systematically incorporate expert judgment

through structured pairwise comparisons, allowing transparent and consistent derivation of relative weights. It is particularly effective in integrating diverse and multi-dimensional datasets such as NDVI, rainfall anomaly, soil moisture, and vegetation stress indices by normalizing them to a common decision scale. AHP provides a flexible, hierarchical decision structure ideal for complex drought modelling problems, where criteria differ in both type and importance. Its consistency check also ensures the reliability of weight assignments, making it a preferred method for spatial drought susceptibility analysis in semi-arid regions.

3.4.1 Weight matrix and pairwise comparison

Using Saaty's scale (Saaty 1980), a pairwise comparison matrix was constructed where each criterion was compared relative to others. The comparative importance is denoted as:

$$P_{ij} = \frac{W_i}{W_j} \tag{9}$$

where, P_{ij} is relative importance of criterion *i* and *j*. W_i and W_j are the weights assigned to criteria *i* and *j*.

3.4.2 Consistency check

Consistency of pairwise judgments was verified using Consistency Index (CI) and Consistency Ratio (CR):

$$CI = \frac{\lambda_{max} - n}{n - 1} \tag{10}$$

$$CR = \frac{CI}{RI} \tag{11}$$

where λ_{max} is the largest eigenvalue and RI is the Random Index (for n=6, RI=1.24).

If CR<0.1, consistency is acceptable (Salomon and Gomes 2024).

3.4.3 Drought Susceptibility Index (DSI)

The normalized thematic layers were integrated into a composite drought susceptibility map using weighted overlay in a GIS platform. The DSI was computed as:

$$DSI = \sum_{i=1}^{n} W_i \,.X_i \tag{12}$$

where W_i is AHP weight of parameter *i* and X_i normalized raster value for parameter *i*.

3.4.4 Weighted Overlay and Mapping

All raster layers were normalized to a common scale and reclassified. The Drought Susceptibility Map (DSM) was created using the equation:

$$DSM = \sum_{i=1}^{n} W_i \,.T_i \tag{13}$$

where W_i is weight from AHP and T_i reclassified raster layer of parameter *i*.

The resulting DSM was categorized into five drought vulnerability classes: very low, low, moderate, high, and extreme. These classifications allow spatial interpretation of drought-prone zones and are critical for mitigation planning and targeted interventions (Li et al. 2023).

4 Results

4.1 Monthly rainfall distribution

Monthly rainfall analysis over the 31-year period (1993–2023) shows that the months from January to May consistently recorded very low precipitation, generally below 10 mm across the Marathwada region (Fig. 3). June marks the beginning of the monsoon, with rainfall increasing from around 50 mm in the eastern districts to more than 180 mm in the western zones. The onset of the monsoon in June creates a distinct spatial disparity, with western districts like Aurangabad and Beed receiving higher early-season rainfall, while eastern districts such as Hingoli and Nanded experience delayed and reduced monsoonal influence. July exhibited the highest monthly rainfall, with values exceeding 300 mm in parts of Aurangabad and Beed. July emerged as the wettest month, with peak rainfall exceeding 300 mm recorded in districts like Aurangabad and Beed, emphasizing the spatial concentration of monsoon intensity in the western part of the region. August followed with moderate rainfall ranging between 180 mm and 250 mm, while September recorded a slightly lower range between 160 mm and 220 mm. October displayed a sharp decline, with rainfall mostly ranging from 20 mm to 60 mm in isolated patches. The gradual decline in rainfall from July through October from over 300 mm to as low as 20 mm, reflects the monsoon withdrawal phase and highlights decreasing water availability for late-season crops. November and December returned to dry conditions, with values again below 10 mm. A pronounced west-to-east rainfall gradient was visible in all monsoon months, particularly June and July, where western districts consistently received higher totals. This west-to-east rainfall gradient, particularly pronounced in

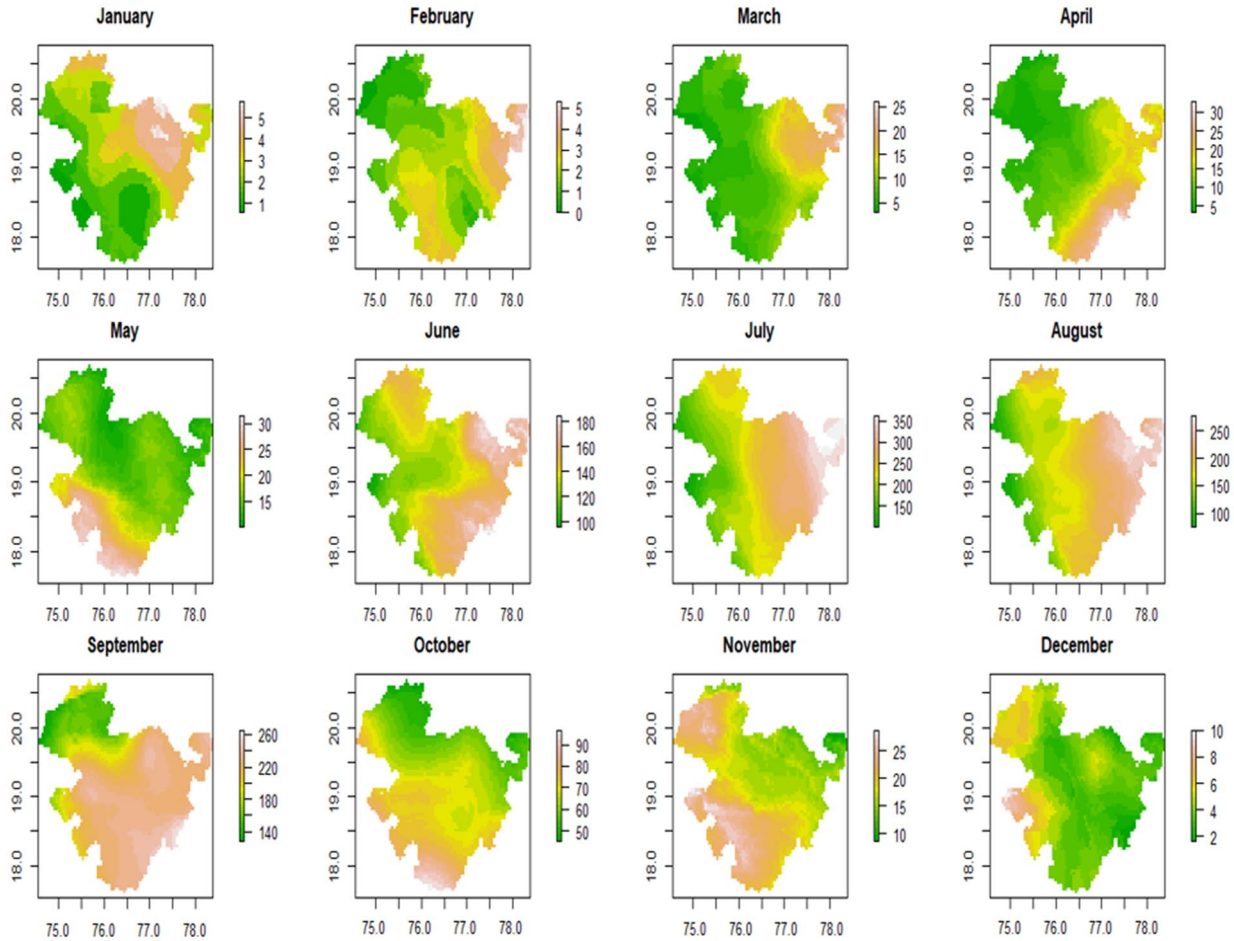


Fig. 3 Map illustrating monthly average rainfall distribution (1993–2023) in Marathwada region

June and July, reflects monsoon wind dynamics and topographic influence, affecting the timing of sowing and irrigation requirements across districts. This long-term analysis highlights a consistent monsoonal concentration of rainfall, with July as the peak month and a pronounced west-to-east gradient in precipitation across the region.

4.2 Seasonal rainfall distribution

Seasonal rainfall maps show that the Pre-Monsoon season (March to May) contributed very little to annual totals, with rainfall across most of the region staying below 40 mm (Fig. 4). Compared to other seasons, the pre-monsoon period (March to May) contributes the least to annual rainfall, with totals below 40 mm, emphasizing Marathwada's heavy reliance on the monsoon (June to September), which supplies nearly 80–90% of total rainfall. The Monsoon season (June to September) was the dominant rainfall period, with cumulative values ranging from 300 mm to 800 mm. The monsoon season (June to September) is the dominant rainfall period, contributing over 80% of the annual precipitation. Western districts like Aurangabad, Beed, and Jalna receive 600 mm to 800 mm, whereas eastern districts such as Hingoli and Nanded get 300 mm to 500 mm. This west-to-east disparity reflects monsoonal wind influence and affects regional agricultural planning. The post-monsoon season (October to November) contributed about 30 mm to 55 mm in central and northern districts. Post-monsoon rainfall (October–November) contributes moderately to seasonal water availability, especially in central and northern districts, where 30–55 mm of rainfall supports late-season crops and soil moisture retention. Although limited, this rainfall also aids shallow groundwater recharge and residual vegetation health before the dry winter months. In contrast, the Winter season (December to February) remained dry, with rainfall across the region consistently below 10 mm. Persistently low winter rainfall (below 10 mm) increases drought vulnerability by restricting soil moisture recharge

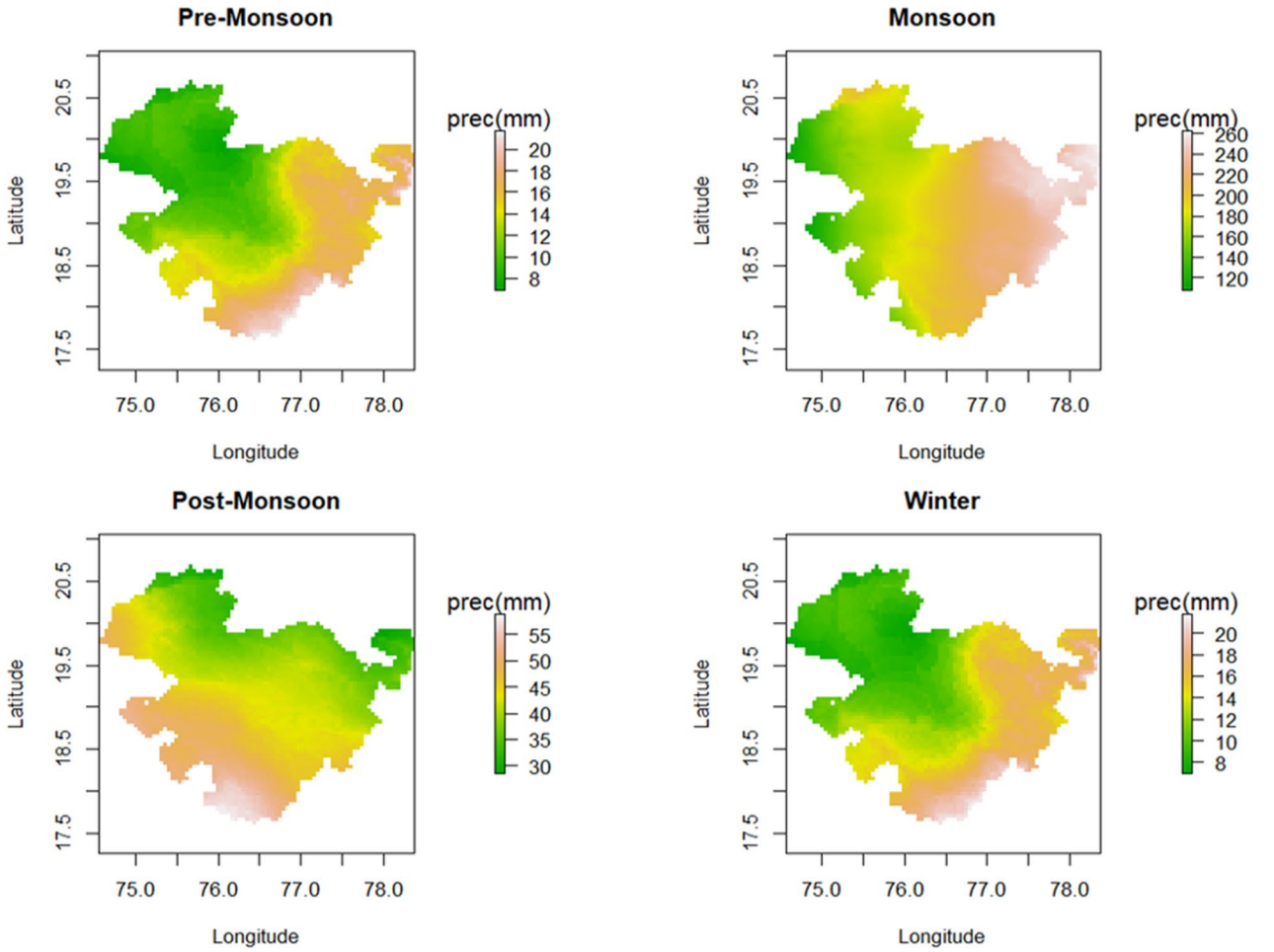


Fig. 4 Map showing seasonal average rainfall distribution (1993–2023) in Marathwada region

and limiting water availability for Rabi crops and groundwater replenishment. This heightens the region's dependence on monsoon rains and underscores the importance of water conservation in drought planning.

4.3 Annual rainfall anomaly and drought analysis

The annual rainfall anomaly analysis indicates that the year 2002 had the most severe negative anomaly, with many districts recording rainfall deficits between −30% and −40% (Fig. 5). Similar patterns were observed in 2004 and 2009, with deficits ranging from −25% to −35%, particularly affecting Parbhani and Hingoli. The year 2012 recorded widespread negative anomalies, ranging from −30% to −50% in several districts, including Aurangabad, Beed, and Latur. The period from 2013 to 2014 showed partial improvement, with anomalies in the range of −10% to +10%. In 2015, negative anomalies once again prevailed, especially in Jalna, Latur, and Parbhani, with values ranging from −25% to −35%. Years such as 2002, 2004, 2009, 2012, and 2015 experienced the most severe negative rainfall anomalies, with Beed, Parbhani, Latur, and Aurangabad among the most affected districts, leading to heightened agricultural stress and water scarcity. During these drought years, rainfall anomalies ranged from −25% to −50%, severely impacting soil moisture recharge, crop yield, and water resources. After 2018, rainfall conditions improved significantly, with 2019 recording a positive anomaly of +32.6%. The trend continued with positive anomalies in 2020 (+31.3%), 2021 (+26.8%), 2022 (+30.3%), and 2023 (+10.5%). This shift in rainfall patterns from severe negative anomalies during earlier drought years to sustained positive anomalies post-2018 reflects improved monsoon performance and partial hydrological recovery in the region. However, it also indicates increasing interannual variability, emphasizing the need for adaptive drought preparedness

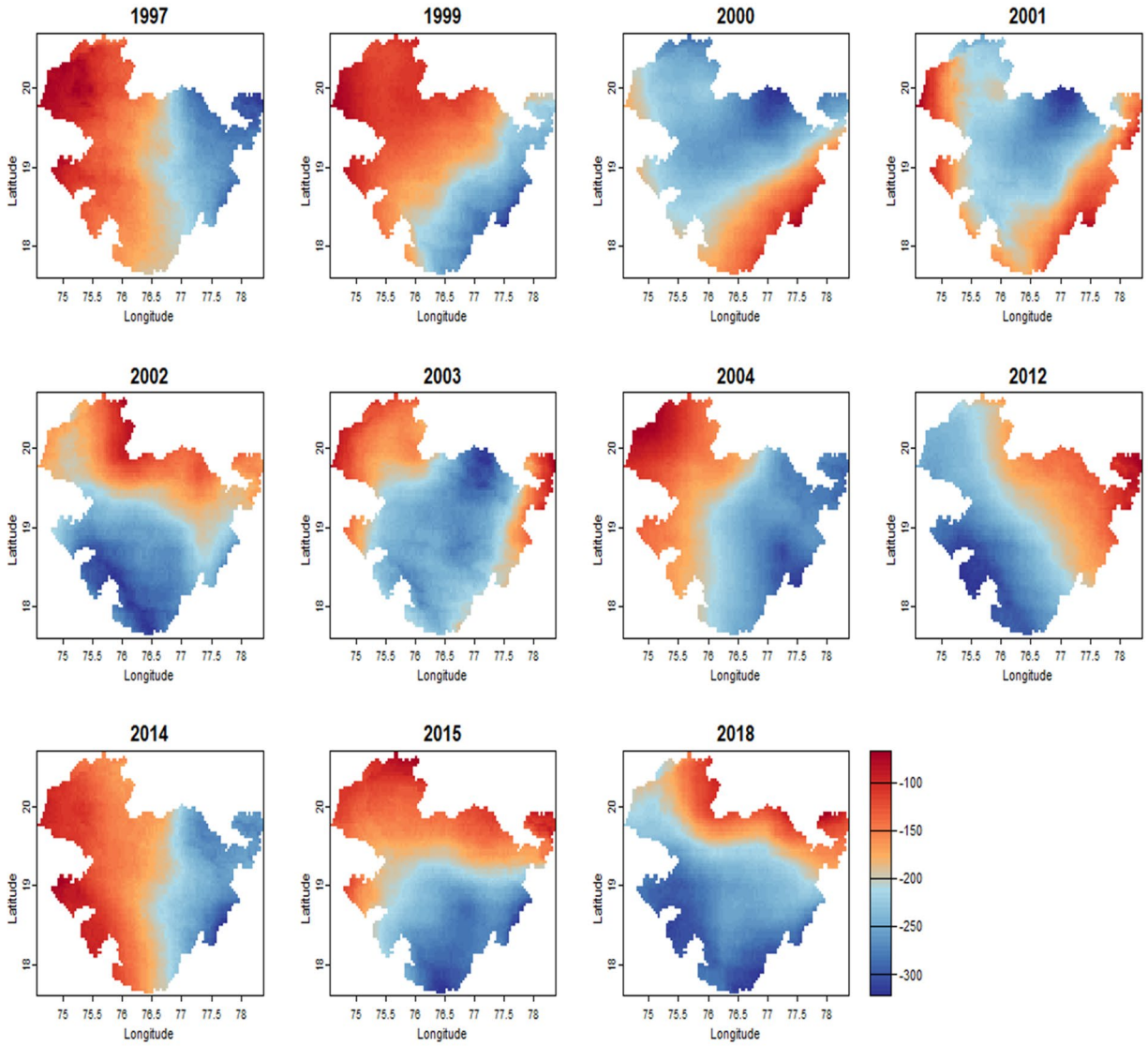


Fig. 5 Map illustrating annual rainfall anomaly for selected drought years

and resilient water resource planning. Long-term rainfall anomaly analysis provides critical insight into the spatial and temporal variability of drought events, enabling early detection of recurring rainfall deficits. This evidence base supports proactive drought mitigation and informs climate-resilient planning in semi-arid regions like Marathwada.

4.4 Annual rainfall deviation

The annual rainfall deviation offers additional insight into drought frequency and intensity (Fig. 6). In 2002, the deviation was −33.7%, while 2004 and 2009 recorded −29.0% and −31.0% respectively. The years 2014 and 2015 showed continued deficits of −19.4% and −24.6%. In contrast, surplus rainfall was noted in 2019 (+32.6%), 2020 (+31.3%), 2021 (+26.8%), 2022 (+30.3%), and 2023 (+10.5%). Several years remained near-normal, including 1995 (–2.3%), 1997 (–1.3%), and 2010 (+0.5%). These statistics confirm a non-linear rainfall pattern across Marathwada. The Marathwada region recorded the highest negative rainfall deviation in 2002 at −33.7%, indicating extreme drought conditions. Conversely, year 2019 experienced the highest positive deviation of +32.6%, reflecting surplus rainfall. These extremes underscore the climatic variability of the region and the need for adaptive water management strategies to address both drought and potential flood risks. Near-normal years, such as 1995, 1997, and 2010, exhibited rainfall deviations within ±5%, indicating relatively stable climatic conditions. These years serve as critical baselines to contrast the impacts of both extreme drought and surplus

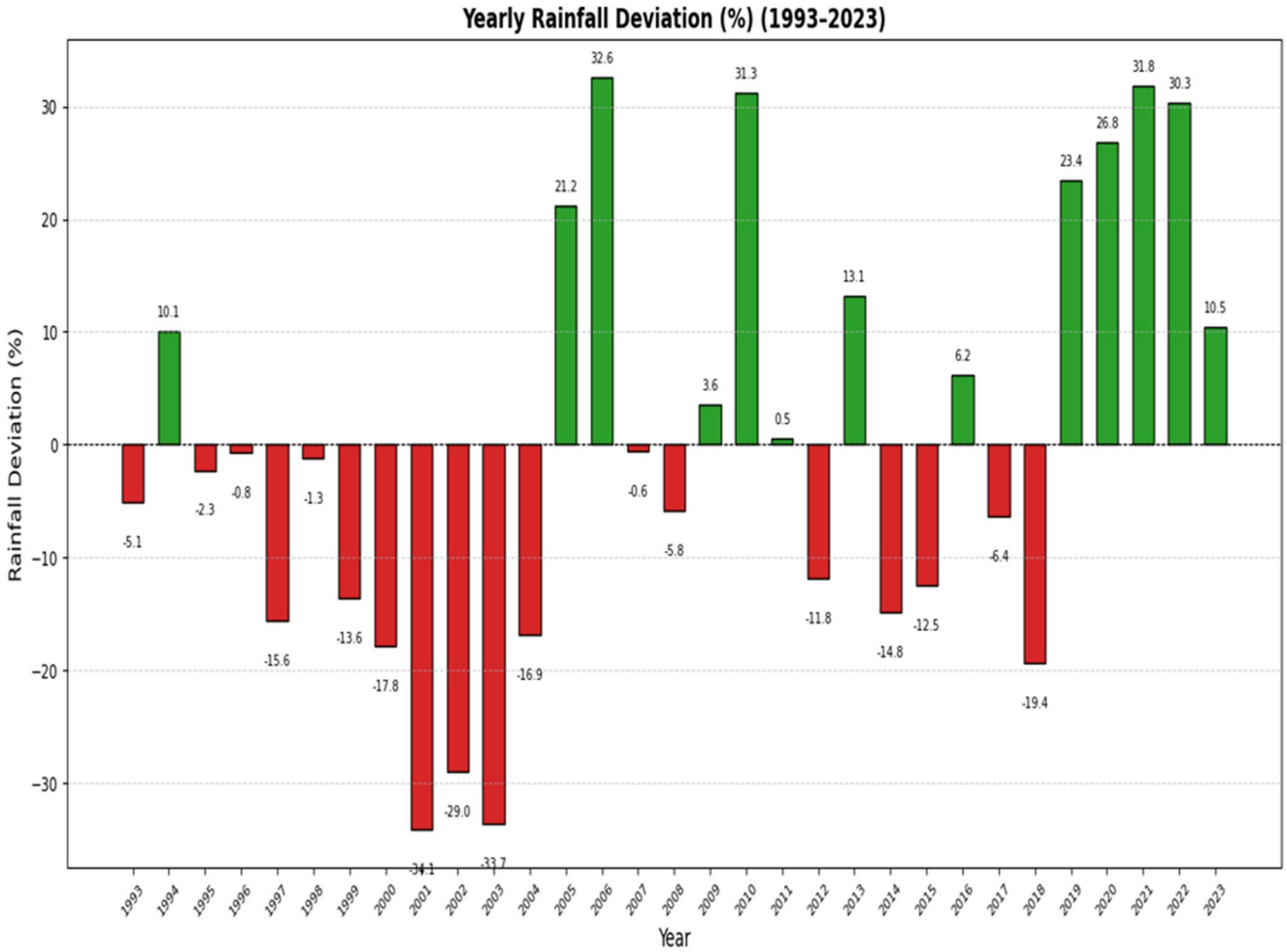


Fig. 6 Graph showing annual rainfall deviation (%) from long-term average (1993–2023)

rainfall years on long-term agricultural and hydrological planning. The observed non-linear pattern in rainfall deviation reflects high interannual variability, which increases uncertainty in monsoon behaviour, complicates crop planning, and elevates drought risk in rainfed agricultural systems. This necessitates the development of climate-resilient water resource and agricultural strategies.

4.5 Agricultural drought using NDVI

The NDVI-based drought analysis showed wide variation across years (Fig. 7). In 2000, NDVI values ranged between −0.1533 and 0.787. Eastern areas showed high vegetation health, while central and southwestern parts had lower values. In 2001, NDVI decreased, ranging from −0.098 to 0.793, with northeastern Marathwada showing more stressed vegetation. The year 2002 recorded one of the worst droughts, with NDVI values between −0.159 and 0.787. Most of the region, especially the central and eastern districts, showed poor vegetation condition. In 2003, NDVI values improved up to 0.822, reflecting better rainfall. NDVI in 2004 also increased, reaching a maximum of 0.814, although western parts still exhibited lower values. In 2012, NDVI reached the highest value of 0.850, suggesting healthy vegetation. The year 2014 saw a steep decline, with NDVI falling to a maximum of 0.524, indicating widespread vegetation stress. In 2018, NDVI ranged from −0.128 to 0.548, showing moderate recovery. During severe drought years such as 2002 and 2014, NDVI values declined substantially across Marathwada, particularly in central and eastern districts. This indicates reduced greenness and heightened vegetation stress resulting from deficient rainfall and soil moisture scarcity. In contrast, years like 2003, 2004, and 2012 showed notable NDVI recovery, aligning with improved rainfall conditions and demonstrating the close link between precipitation and vegetation health. Eastern regions, especially Hingoli and Nanded, consistently maintained higher NDVI values even during moderate drought years, suggesting spatial heterogeneity in vegetation resilience.

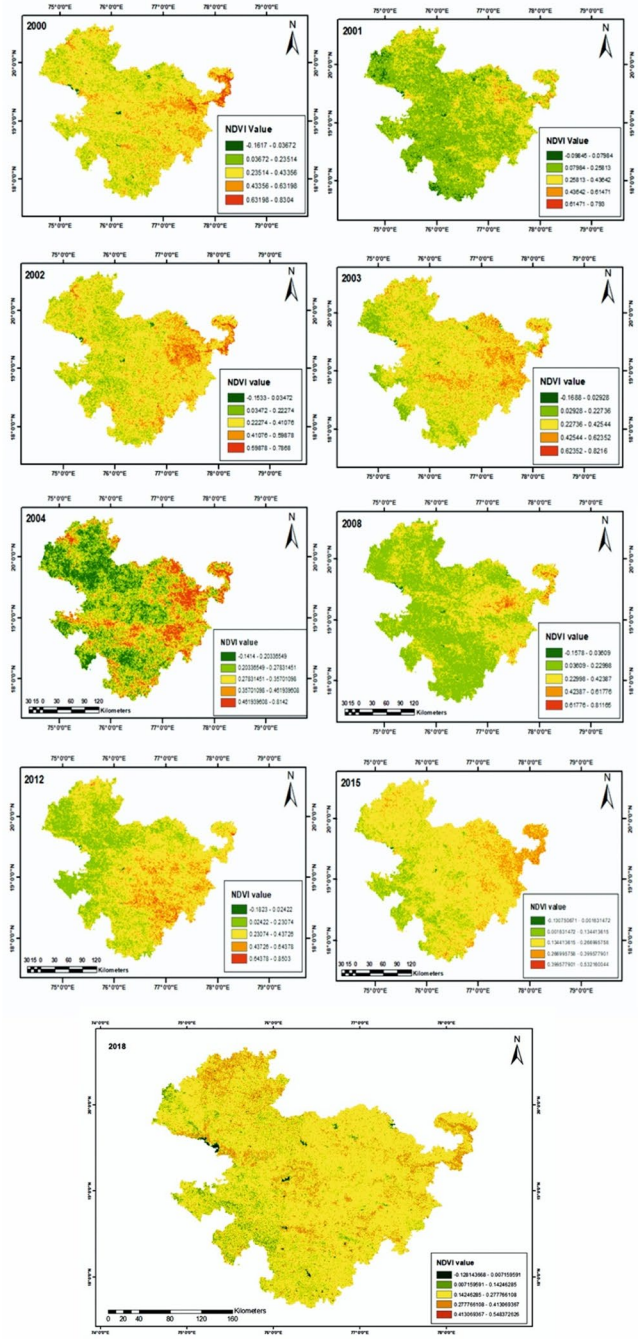


Fig. 7 NDVI maps illustrating vegetation health across Marathwada for key drought years

4.6 Temperature condition index (TCI) evaluation

The Temperature Condition Index (TCI) maps displayed strong interannual variation (Fig. 8). In 2000, moderate to high TCI values were observed across the central region, while the southern border showed low TCI, indicating thermal stress. The year 2001 reflected lower TCI in the northern and eastern districts. In 2002, TCI improved in central parts, indicating partial recovery. In 2003, TCI declined again, especially in the southern and western regions. The year 2004 showed higher TCI values in southern districts, while the north remained stressed. In 2012, extensive low TCI values were noted in the northeast and centre, reflecting surface temperature stress. The years 2014 and 2015 had large zones of low TCI in the northwest, west, and south. TCI values in 2018 ranged widely (0–100), but large grey zones indicated missing or non-vegetated areas.

4.7 Vegetation condition index (VCI) analysis

The VCI data also revealed significant drought trends (Fig. 9). In 2000, most areas had VCI between 30 and 100, with minimal stress. A sharp decline was seen in 2001, with southern and eastern parts showing VCI below 20. In 2002, stress continued in the north-central zones, but some improvement was noted in the south and west. VCI in 2003 dropped again, with severe stress in southern and western districts. In 2004, vegetation health improved moderately, although eastern regions still showed VCI below 30. The year 2008 experienced low VCI across most of Marathwada, confirming drought conditions. In 2012, northeast and southwest districts had VCI values in the 0–20 range, showing severe stress. In 2015, VCI dropped further, with central and southern Marathwada showing widespread degradation. By 2018, vegetation conditions had partially improved, with most regions recording VCI in the 30–50 range, though pockets of stress remained.

4.8 Vegetation health index (VHI)

The Vegetation Health Index (VHI) analysis from 2000 to 2018 shows clear spatial and temporal drought patterns in Marathwada (Fig. 10). In 2000, over 80% of the area showed healthy conditions with VHI above 40, indicating minimal drought. However, in 2001, more than 60% of the region fell below a VHI of 20, with Latur, Jalna, and Parbhani showing severe to extreme stress below 10. The drought continued in 2002, affecting around half the region, especially in Jalna, Parbhani, and Nanded with VHI between 10 and 30. In 2003, over 50% of the area recorded VHI less than 30, with Beed and southern Latur showing critical values under 10. Some recovery occurred in 2004 with moderate VHI values (30–50) in central and western zones, although Jalna remained under stress. In 2008, over 40% of the land again showed VHI under 30, particularly in Nanded and Parbhani. Severe drought returned in 2012, with VHI values under 20 in more than 50% of the area. The most extreme conditions were recorded in 2015 when about 75% of Marathwada had VHI values under 20, and districts like Beed and Latur had vast areas under VHI < 10. In 2018, partial recovery was visible with several central regions showing VHI above 40, but southern Beed and Osmanabad continued to face stress with VHI under 30.

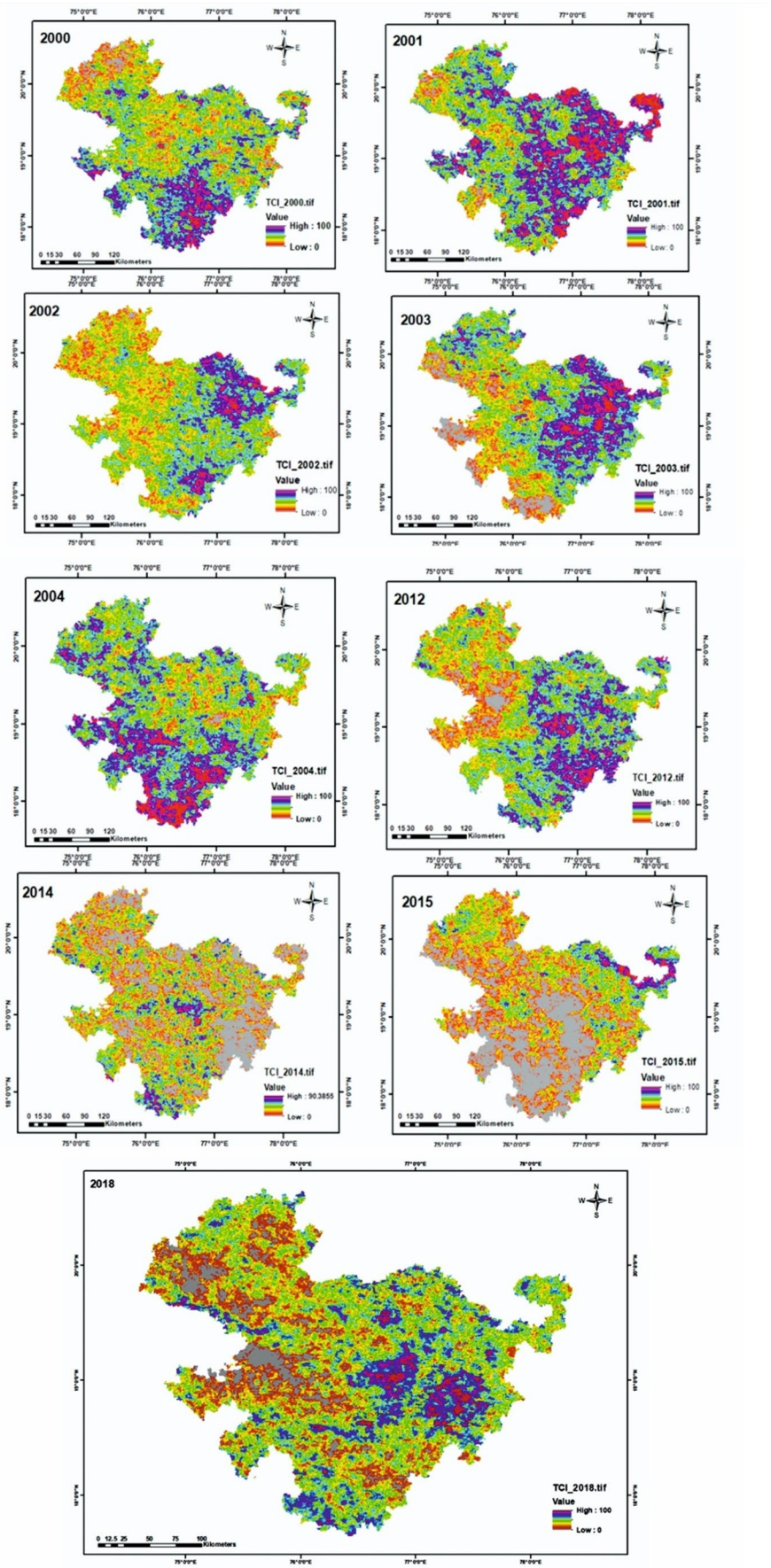


Fig. 8 TCI maps for Marathwada showing spatial distribution of thermal stress during major drought years

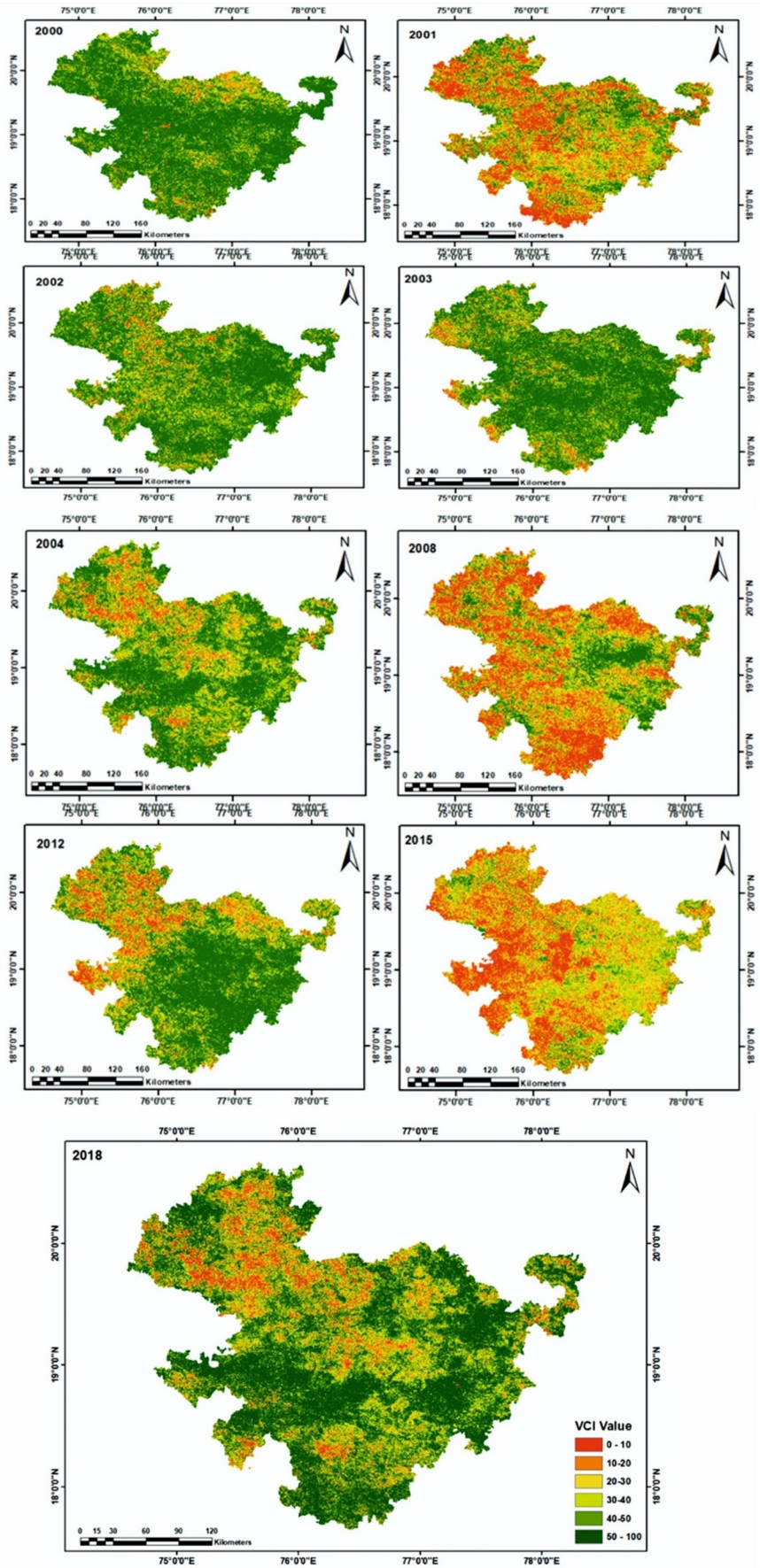


Fig. 9 VCI maps for Marathwada showing spatial vegetation stress levels during drought years

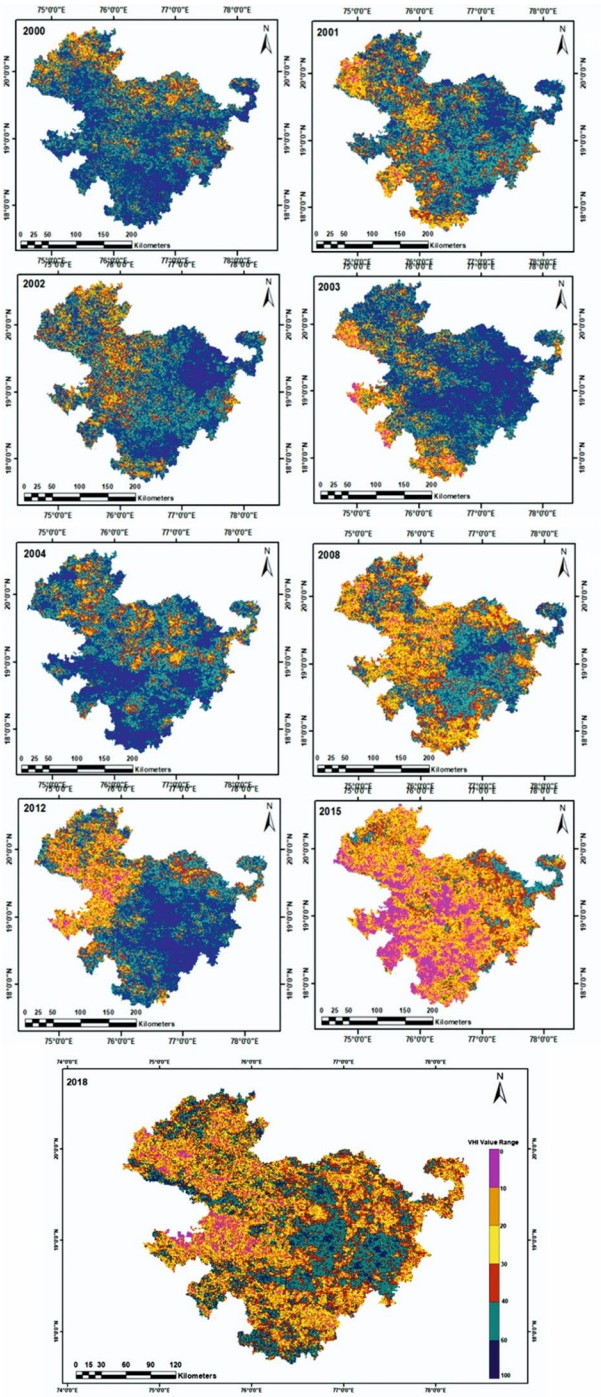


Fig. 10 Maps showing Vegetation Health Index (VHI) for drought years across Marathwada

4.9 Soil moisture anomaly percentage index (SMAPI)

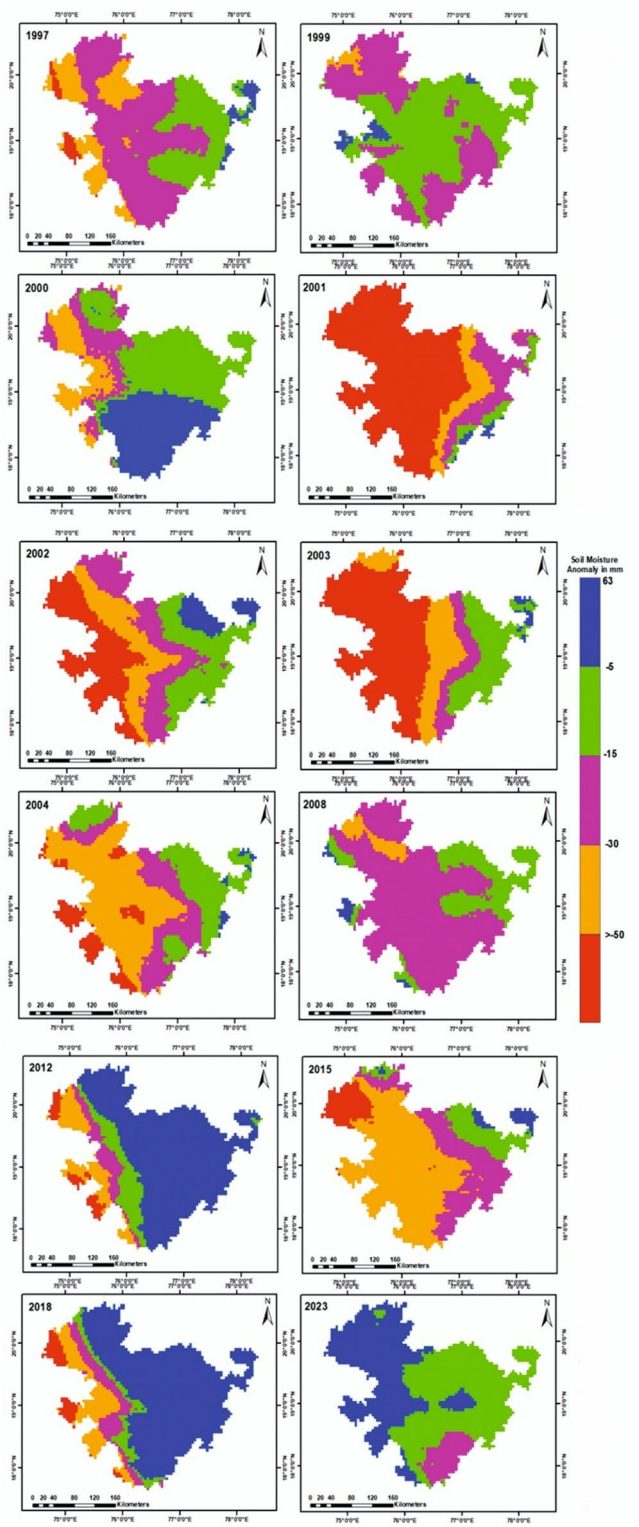


Fig. 11 Maps illustrating SMAPI for drought years across Marathwada

The SMAPI results between 1997 and 2023 further confirmed severe soil moisture stress during drought years (Fig. 11). In 1997 and 1999, over 60% of the region recorded anomalies below −30 mm. In 2001, nearly 80% of Marathwada showed SMAPI under −50 mm, indicating critical soil water loss, especially in Beed and Osmanabad. The drought persisted through 2002 and 2003 with over 70% of western and central districts falling under SMAPI

< −30 mm. Moderate deficits ranging from −15 mm to −30 mm were seen in southern areas in 2004 and 2008. A pronounced deficit reappeared in 2012 when more than 50% of southern Marathwada recorded anomalies below −30 mm, particularly in Latur and Hingoli. The most severe SMAPI drop was again seen in 2015, with Beed and Parbhani registering anomalies below −50 mm. In 2018, western parts of Marathwada exhibited significant negative values, indicating poor moisture conditions. The year 2023 saw overall better conditions, although southeastern districts like Latur and Parbhani still recorded SMAPI below −30 mm.

4.10 Soil moisture anomaly (SMA)

The long-term Soil Moisture Anomaly (SMA) trend from 1993 to 2023 revealed strong interannual variability (Fig. 12). Negative SMA values were recorded in 17 out of the 31 years. The most severe drought year was 2002, with SMA reaching −24.0 mm. Other critical years included 2001 and 2003, both recording anomalies below −15 mm. A prolonged dry period persisted from 2001 to 2005. In 2015, SMA dropped to −13.9 mm, confirming widespread soil moisture deficit. On the other hand, 13 years showed positive anomalies. The year 2011 recorded the highest positive SMA of +14.8 mm, followed by 2010 and 2013 with values exceeding +10 mm. Wet years such as 1994, 1997, and 2020 also showed improved soil conditions. In 2023, the SMA hovered around −2.5 mm, suggesting near-normal soil moisture but with slight deficits in localized areas. The long-term SMA trend reveals sharp interannual variability in soil moisture across Marathwada, reflecting the region's sensitivity to rainfall anomalies. Severe negative anomalies in 2002, 2001, and 2015 indicate prolonged drought stress, disrupting crop cycles and delaying hydrological recovery. In contrast, positive SMA years like 2010, 2011, and 2013, driven by above-average rainfall, enhanced groundwater recharge and vegetation health. The near-normal SMA in 2023 (–2.5 mm) suggests stable moisture conditions conducive to regular agricultural planning. These fluctuations underscore the importance of continuous SMA monitoring for drought preparedness and adaptive resource management.

4.11 Drought susceptibility modelling

A spatially integrated drought susceptibility model was developed using Analytic Hierarchy Process (AHP) in a GIS environment. Six critical parameters were used: NDVI, Rainfall Anomaly, SMAPI, VCI, VHI, and TCI. These were selected based on their scientific relevance in assessing drought conditions in semi-arid regions (Fig. 13a-f). The six parameters represent key climatic and biophysical indicators critical for assessing drought conditions. Rainfall anomaly identifies long-term precipitation deficits, while SMAPI captures soil moisture availability, crucial for crop sustainability. NDVI and VCI reflect vegetation health and

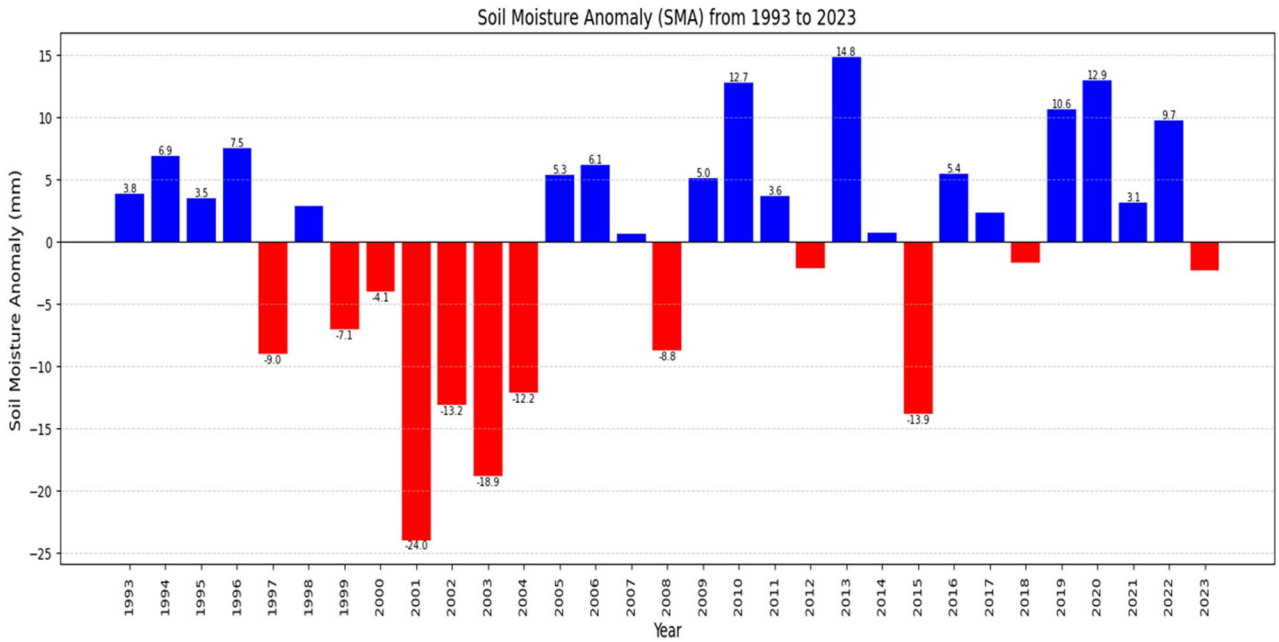


Fig. 12 Bar diagram showing Soil Moisture Anomaly (SMA) from the year 1993 to 2023

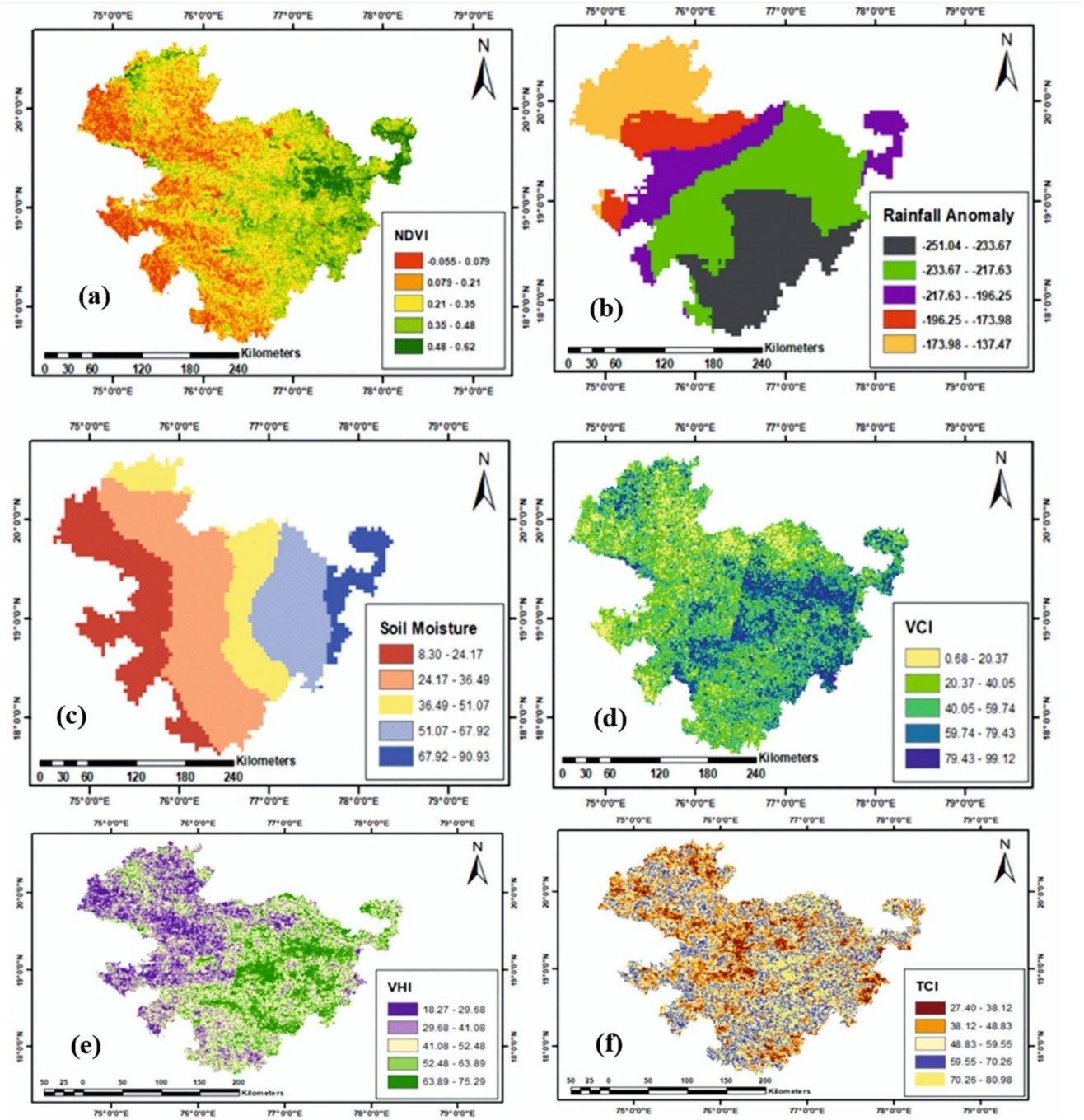


Fig. 13 Map illustrates six spatial layers such as NDVI (13a); Rainfall Anomaly(13b); Soil Moisture(13c); VCI (13d); VHI (13e); and TCI (13f), used in the AHP-based weighted overlay analysis to derive drought susceptibility zones across the Marathwada region

greenness, indicating plant stress levels. VHI combines vegetation and temperature data, while TCI highlights thermal stress. Together, they offer a comprehensive understanding of agricultural drought severity across temporal and spatial scales.

NDVI ranged from −0.05 to 0.62. Eastern districts like Hingoli, Nanded, and Parbhani exhibited higher NDVI values (0.35–0.62), indicating healthy vegetation. Central and western zones had lower NDVI values (<0.21), showing vegetation stress. Rainfall anomaly values ranged between −251.04 mm and −137.47 mm over the 30-year dataset (1993–2023). Beed, Osmanabad, and Latur recorded deficits exceeding −230 mm, placing them in Extreme Susceptibility class. Soil moisture values varied from 8.30% to 90.93%. Eastern Marathwada recorded higher moisture (>68%) especially in Hingoli and Nanded, while the

western districts showed low moisture (<36%), indicating high drought stress. VCI values ranged from 0.68 to 99.12. Higher VCI values (>74) were recorded in eastern Parbhani and Hingoli, indicating robust vegetation. Lower values (<30) in western zones suggested crop stress. VHI ranged between 18.27 and 75.29. Poor vegetation health (<41.08) was observed in southwest and northwest Marathwada, correlating with high susceptibility. TCI values varied from 27.40 to 80.96. Low values (<48.83) were mostly observed in the central and southern areas, reflecting thermal stress. Higher TCI (>70) in northeast indicated favourable thermal conditions.

4.12 Pairwise comparison matrix and parameter weights

To assign weights, expert judgement was applied using a pairwise comparison matrix (Table 2). The consistency ratio (CR = 0.031) was within acceptable limits (<0.10), validating the matrix consistency. The Rainfall Anomaly received the highest weight (27.44%) followed by VHI (28.42%), Soil Moisture (25.59%), VCI (8.97%), NDVI (6.47%), and TCI (3.11%). Rainfall Anomaly received the highest weight (27.44%) in the AHP analysis. It is considered crucial because rainfall deficiency is the primary trigger for drought in semi-arid regions like Marathwada. Prolonged negative anomalies directly affect soil moisture, vegetation health, and crop productivity. Thus, its influence is fundamental in identifying and predicting drought-prone zones.

4.13 Drought susceptibility map

The drought susceptibility map, developed through AHP-based weighted overlay of six parameters, reveals clear spatial variation across the Marathwada region (Fig. 14). Areas classified under Extreme and Severe drought susceptibility are located mainly in Beed, Osmanabad, Latur, and parts of Jalna and Parbhani, where rainfall anomalies exceed −230 mm, soil moisture levels fall below 36%, VHI values remain under 41%, and TCI drops below 48%. These districts are further characterized by low NDVI and VCI values, reflecting stressed vegetation, and high surface temperature stress indicated by TCI values below 48%. Collectively, these hydro-climatic and vegetation indicators highlight the critical vulnerability of these zones. In contrast, Moderate drought susceptibility is observed across central Parbhani, Aurangabad, and mid-Nanded, where rainfall deficits range from −217 mm to −196 mm, soil moisture levels vary between 36% and 67%, NDVI values lie between 0.21 and 0.35, VCI ranges from 40 to 60, and TCI is between 48% and 59%. These areas exhibit intermediate drought risk, marked by partial moisture retention and moderate vegetation health, distinguishing them from both highly stressed and resilient zones. The Low and Very Low susceptibility zones are predominantly located in Hingoli, eastern Nanded, and portions of Parbhani and Jalna, characterized by rainfall anomalies between −196 mm and −137 mm, soil moisture exceeding 68%, NDVI above 0.35, VCI above 60, VHI exceeding 63%, and TCI above 70%. These districts demonstrate favourable hydro-climatic and ecological conditions, indicating better vegetation stability and lower drought exposure. The classification implies stronger climatic resilience and more sustainable agricultural conditions in these regions. These regions indicate relatively stable climatic and vegetation conditions and lower drought vulnerability. The drought susceptibility map was classified using AHP-based weighted overlay of six parameters into five categories: Very Low, Low, Moderate, Severe, and Extreme. Classification was based on cumulative scores derived from parameter weights and normalized spatial raster values. This categorization helps distinguish areas with varying drought risk, enabling targeted interventions. It supports efficient planning for drought mitigation, agricultural adaptation, and water resource management in vulnerable semi-arid zones. A summary table was developed to link major drought years with dominant stress indicators and affected districts across Marathwada (Table 3).

5 Discussion

The present study reveals strong seasonal and spatial variability in rainfall and drought susceptibility across the Marathwada region. The spatial and temporal rainfall trends from 1993 to 2023 show a strong west-to-east gradient, with western districts receiving higher monsoon rainfall.

Table 2 Pairwise comparison matrix of selected parameters for drought susceptibility modelling

Criteria	Rainfall Anomaly	VHI	Soil Moisture	VCI	NDVI	TCI	Weight (%)
RA	1	1	1	5	5	5	27.44
VHI	1	1	1	5	5	7	28.42
SM	1	1	1	3	5	7	25.59
VCI	1/5	1/5	1/3	1	1	7	8.97
NDVI	1/5	1/5	1/5	1	1	3	6.47
TCI	1/5	1/7	1/7	1/7	1/3	1	3.11
				$\lambda_{max} = 6.152$	**CI = 0.038**	**CR = 0.031**	**CR < 0.1**

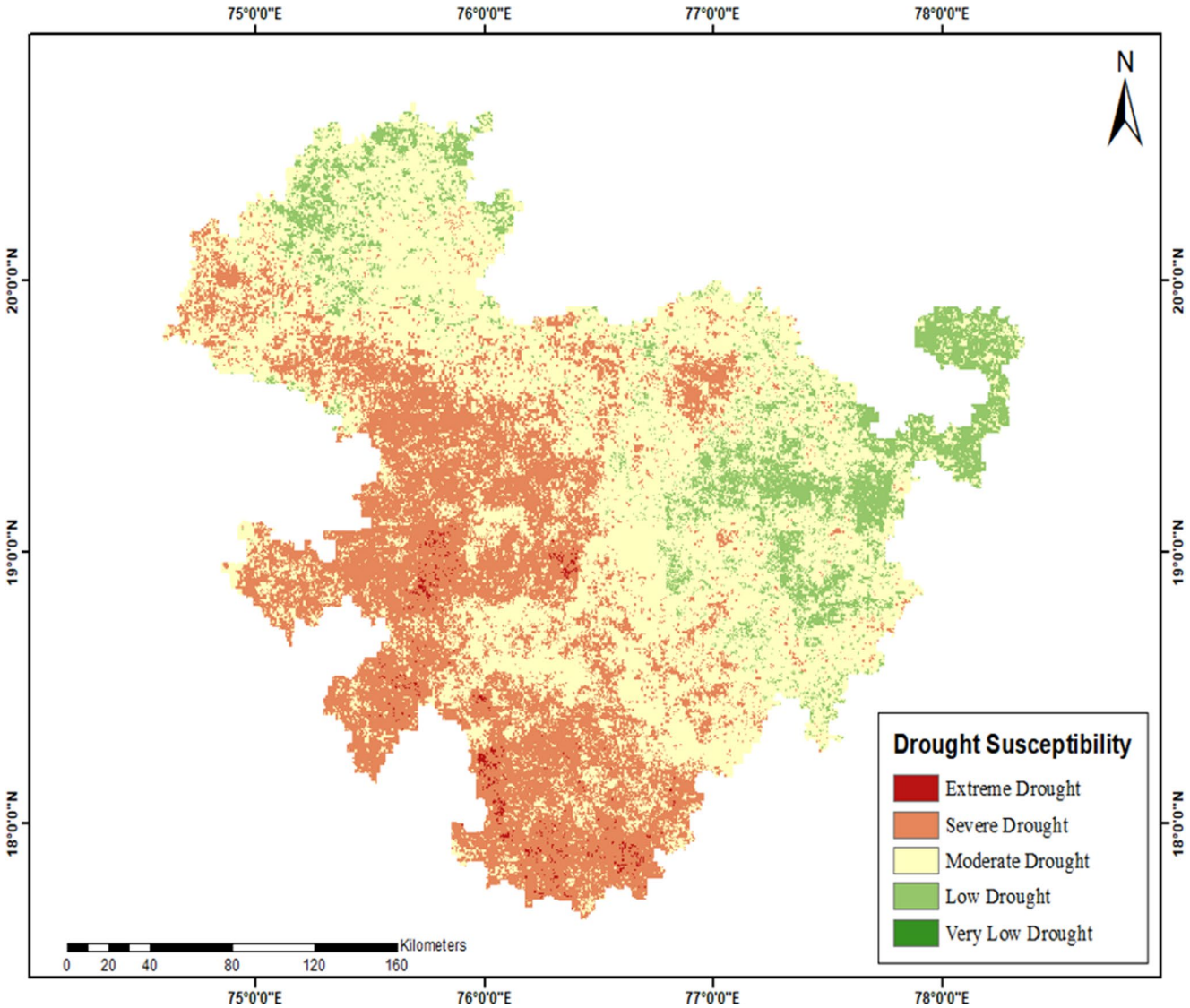


Fig. 14 Drought susceptibility map of Marathwada showing spatial distribution of drought vulnerability derived from AHP-integrated weighted overlay of six key environmental parameters

Table 3 Summary of major drought years with dominant stress indicators and affected districts

Year	Dominant indicator(s)	Affected districts
2002	Rainfall anomaly, NDVI, VHI, SMAPI	Beed, Parbhani, Hingoli, Jalna, Nanded
2004	Rainfall anomaly, NDVI, TCI	Jalna, Parbhani, Hingoli
2009	Rainfall anomaly	Hingoli, Nanded
2012	NDVI, VCI, VHI, SMAPI	Aurangabad, Beed, Latur, Parbhani, Hingoli
2015	VHI, TCI, SMAPI, NDVI	Beed, Latur, Jalna, Parbhani, Osmanabad
2018	NDVI, VCI, SMAPI	Osmanabad, Jalna, Latur
2023	SMAPI, SMA (localised moisture deficit)	Latur, Parbhani

Years such as 2002, 2004, and 2015 exhibited severe rainfall deficits, with deviations ranging from −25% to −40%. These findings are aligned with previous studies on repeated drought events in semi-arid parts of Maharashtra (Amrit et al. 2020; Singh et al. 2021). The NDVI, VCI, and VHI indices during drought years revealed considerable vegetation stress in districts like Beed, Latur, and Jalna. Vegetation indices such as NDVI, VCI, and VHI played a crucial role in identifying agricultural stress during severe drought years like 2002 and 2015, with values falling below critical thresholds (NDVI < 0.2, VCI and VHI < 30), reflecting widespread crop deterioration and confirming the intensity of agricultural drought across Marathwada. TCI and

SMAPI complement rainfall data by capturing additional dimensions of drought stress. Low TCI values (below 48%) indicate elevated land surface temperatures that accelerate evapotranspiration, while SMAPI values below −30 mm reflect significant soil moisture deficits. Together, these indicators provide a more comprehensive understanding of drought severity, especially in semi-arid regions where temperature and soil moisture play critical roles in agricultural vulnerability. In 2015, VHI dropped below 20 in over 75% of the region, aligning with earlier findings on extreme vegetation degradation during drought years (Mullapudi et al. 2023). These results suggest that recurring monsoon failure, soil moisture loss, and high temperatures collectively lead to severe agricultural stress in central Marathwada. Although this study focuses on historical drought susceptibility, the observed increase in drought frequency and intensity over the 31-year period may reflect broader climate change impacts. Rainfall anomaly and soil moisture are key indicators of agricultural drought in semi-arid regions. A significant rainfall deficit leads to reduced infiltration and poor soil moisture retention, limiting water availability for crops. Low soil moisture directly stresses plant roots, reducing growth and yield. Together, these factors intensify drought severity and increase vulnerability in rain-fed farming systems. The main causes of drought vulnerability in the semi-arid Marathwada region include highly variable and declining monsoon rainfall, with anomalies reaching up to −251 mm in certain years. Low soil moisture levels (<36%) and poor vegetation health (NDVI<0.21, VHI<41%) further reduce agricultural resilience. High land surface temperatures (TCI<48%) intensify evapotranspiration, worsening water stress. These climatic and biophysical factors, combined with limited irrigation infrastructure, make the region highly susceptible to recurring droughts.

The drought susceptibility modelling using AHP provided a comprehensive spatial picture of drought-prone zones. Extreme and Severe classes were found predominantly in Beed, Osmanabad, and Latur, where rainfall anomalies exceeded −230 mm and soil moisture dropped below 36%. These results are consistent with those reported by Ganjir et al. (2025), who also identified these districts as drought hotspots. Moderate zones appeared in Aurangabad and Parbhani, where indicators were relatively balanced. Low and Very Low drought zones, such as Hingoli and eastern Nanded, had higher NDVI (>0.35), VHI (>63%), and SMAPI (>68%). The AHP model effectively prioritized parameters based on expert consensus, with Rainfall Anomaly (27.44%) and VHI (28.42%) receiving the highest weights. Similar weighting strategies have been applied in other regional drought models (Palchaudhuri and Biswas 2016; Alharbi et al. 2022). The use of satellite-derived environmental indicators added accuracy to the spatial classification, especially when assessing both climatic and vegetation-based stress signals together (Zeng et al. 2023; Leisenheimer et al. 2024).

The main advantage of this study is the integration of several drought indices using AHP in a GIS platform. It offers a structured and objective way to rank vulnerability based on six parameters, producing a detailed spatial output useful for local planning. Unlike single-index models, it captures both meteorological and agricultural drought dimensions (Baig et al. 2020). However, limitations exist. The model depends on expert judgement, and satellite data may miss ground realities, such as localized irrigation or cropping intensity. The model excludes critical factors like groundwater depletion and socio-economic vulnerability due to data constraints. Seasonal cloud cover also affects NDVI accuracy (Dhawale and Paul 2018; Essaadia et al. 2022). In addition, the use of varying spatial and temporal resolutions in satellite products introduces uncertainty, which future studies should quantify through validation. Primary data collection through farmer surveys and participatory rural appraisals can improve local validation and adaptation planning (Senapati and Das 2025; Senapati et al. 2025b). Despite these gaps, the model offers a reliable drought mapping approach. However, this study was not validated against observed impacts such as crop yield loss, historical drought years, government drought declarations, or groundwater depletion data, which limits operational validation. Satellite-derived indices may be affected by cloud cover, resolution limitations, and vegetation type variability, which can reduce accuracy in assessing ground conditions. The AHP method relies on expert judgment, which introduces subjectivity and potential bias in parameter weighting. These factors may limit the precision of drought classification. Future studies could integrate ground-truth data, machine learning techniques, and real-time sensor networks to enhance model reliability. In future research, the model can be enhanced using socio-hydrological indicators, high-resolution temporal data, and dynamic machine learning approaches to predict evolving drought conditions in real-time (Senapati and Das 2022; Senapati et al. 2023b; Kartal et al. 2024; Azmir 2025). Such integrated efforts will better inform drought contingency frameworks and support sustainable agricultural resilience in semi-arid regions. Moreover, incorporating land-use and ecological modelling can guide adaptive agricultural practices, crop diversification, and long-term land planning under drought stress (Senapati et al. 2023a, 2024; Senapati and Das 2024). The spatial multicriteria drought assessment provides detailed maps that identify zones with varying drought risk levels. Government agencies can use this information for targeted drought mitigation, resource allocation, and crop planning. Local planners can prioritize vulnerable districts like Beed and Latur for irrigation development and contingency

planning. This approach supports proactive decision-making to enhance drought resilience at regional and local levels. Integrating future climate projections could further enhance the applicability of this model for long-term agricultural resilience planning.

6 Conclusions

This study applied an integrated AHP–GIS framework to map drought susceptibility across the semi-arid Marathwada region using key hydro-climatic and vegetation indicators. By combining rainfall anomaly, soil moisture, NDVI, VCI, VHI, and TCI, the model captured both meteorological and agricultural dimensions of drought risk. The results revealed significant spatial disparities: districts such as Beed, Latur, and Osmanabad were classified as highly susceptible, while areas like Hingoli and eastern Nanded showed greater resilience due to more favourable climatic and vegetation conditions. The use of satellite-based indices enhanced the model's ability to monitor real-time drought dynamics, offering timely and location-specific insights. This approach provides a practical decision-support tool for policymakers and planners, enabling targeted drought mitigation, sustainable water management, and agricultural resilience. The methodology is cost-effective, adaptable, and scalable, making it applicable to other drought-prone regions globally under changing climate scenarios.

7 Research highlights

- A geospatial drought susceptibility model was created using AHP and six key environmental indicators.
- Rainfall anomaly (−251 mm) and soil moisture (<36%) were major contributors to extreme drought zones.
- Final map categorized Marathwada into five drought risk zones: very low to extreme.

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Author contributions KJ: designed research, visualization, writing original manuscript and editing, supervision. SG: collected data, analysed data, contributed to software, data support, conceived, writing original manuscript. SKS : conceived, designed research, supervision, analysed data, writing original manuscript and editing. All authors reviewed the manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethical approval All authors have read, understood, and have complied as applicable with the statement on "Ethical responsibilities of Authors" as found in the Instructions for Authors.

Consent to participate Agree to participate.

Consent to publish Agree to publish.

Competing interests The authors declare no competing interests.

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